

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

Kopiereg voorbehou

Semester Test 1

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Module EBN111

Elektrisiteit en Elektronika

6 March 2010

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Electricity and Electronics

6 March 2010



UNIVERSITEIT VAN PRETORIA
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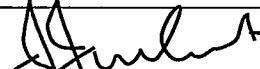
Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute 90 minutes	Oopboek / toeboek: <i>Open / closed book:</i>	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	18		

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert		
	2 DV Bhatt		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>			
Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	

VRAAG/QUESTION 1 [12]

Vraag 1.1 [1]

'n Herlaaibare flitsligbattery kan 50mA stroom lewer vir 10 ure met 'n terminaalspanning van 1.5V. Hoeveel energie kan die battery lewer?

$$I = 50 \text{ mA}$$
$$t = 10 \text{ h}$$
$$V = 1.5 \text{ V}$$

Question 1.1 [1]

A rechargeable flashlight battery is capable of delivering 50mA for 10 hours at a terminal voltage of 1.5V. How much energy can the battery deliver?

$$W = P \cdot t = V \cdot I \cdot t$$
$$= 1.5 \cdot 50 \text{ mA} \cdot 10 \text{ h}$$
$$= 0.75 \text{ Wh}$$
$$= \underline{2.71 \text{ kJ}} \quad \checkmark \quad (1)$$

Vraag 1.2 [1]

Centurion munisipaliteit vra 75 sent/kWh. A sekere kantoorgebou gebruik 30 gloeilampe vir 31 dae lank (5 ure per dag). Bereken die elektrisiteitsrekening as 100W gloeilampe gebruik word.

Question 1.2 [1]

Centurion municipality charges 75 cents/kWh. A certain office building uses 30 bulbs for 31 days (5 hours per day). Determine the electricity bill if the rating on the bulbs is 100W.

$$\text{Bill} = 100 \text{ W} \times 30 \times 31 \text{ days} \times 5 \text{ h} \times 75 \text{ c/kWh}$$
$$= 348750 \text{ Wh/kWh} = \underline{R348.75} \quad \checkmark \quad (1)$$

Vraag 1.3 [2]

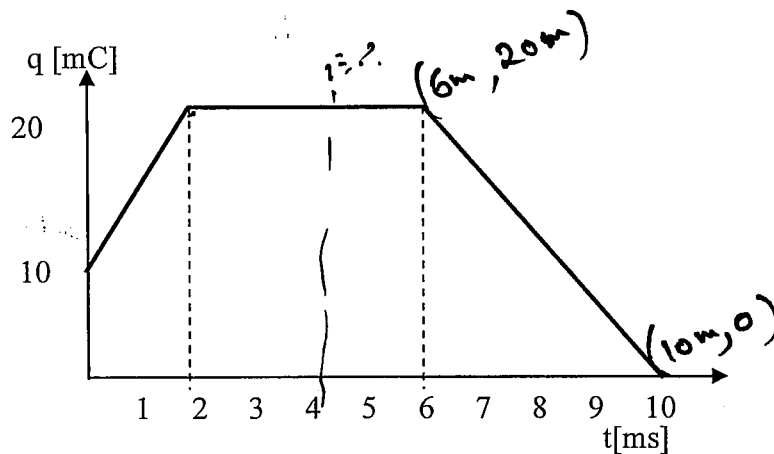
Lading wat deur 'n element vloei is in die grafiek geplot.

- a) Lei 'n vergelyking af vir lading as funksie van tyd vir $6\text{ms} \leq t \leq 10\text{ms}$.
 b) Bereken die stroom by $t = 4\text{ms}$.

Question 1.3 [2]

Charge flowing through an element is plotted in the graph.

- a) Derive an expression for the charge as a function of time for $6\text{ms} \leq t \leq 10\text{ms}$.
 b) Calculate the current at $t = 4\text{ms}$.



$$a) (q - 0) = \frac{0 - 20\text{m}}{10\text{m} - 6\text{m}} (t - 10\text{m}) \quad y - y_2 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_2)$$

$$q = \frac{-20\text{m}}{4\text{m}} (t - 10\text{m})$$

$$q = -5 (t - 10\text{m}) \quad \text{C.} \rightarrow \textcircled{1}$$

$$b) i = \frac{\Delta q}{\Delta t} = \frac{20\text{m} - 20\text{m}}{4\text{m} - 2\text{m}} = \frac{0}{2\text{m}} = 0\text{A} \rightarrow \textcircled{1}$$

Vraag 1.4 [4]

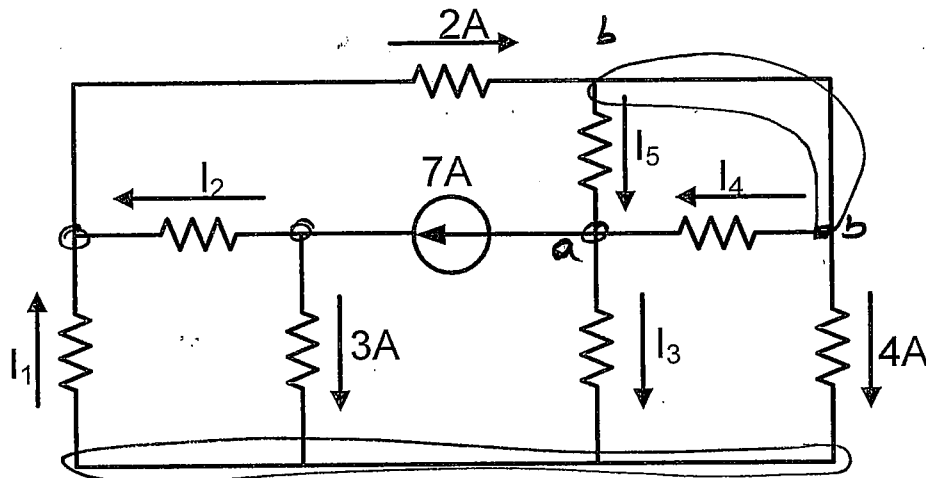
Bepaal

- a) Die aantal takke en nodes.
b) I_3 .

Question 1.4 [4]

Find

- a) The number of branches and nodes.
b) I_3 .



Branch = 9 ✓
Nodes = 5 ✓

I_3 : $-7 + I_5 + I_4 + I_3 = 0$
 $\therefore I_3 = -7 + (I_4 + I_5)$ but

but

$$2 - I_4 - I_5 - 4 = 0$$

$$\therefore I_4 + I_5 = -2$$

$$\therefore I_3 = -7 - 2 = -9A$$

②

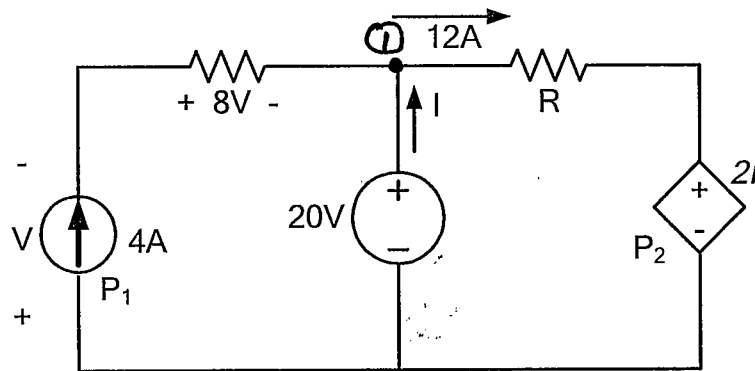
or

Vraag 1.5 [4]

Bereken V , I en die drywings P_1 en P_2 .

Question 1.5 [4]

Calculate V , I and the powers P_1 and P_2 .



KVL:

$$+V + 8 + 20 = 0$$

$$\therefore V = -28V \checkmark \textcircled{1}$$

$$P_1 = +i \cdot v$$

$$= 4 \cdot -28$$

$$= -112W \checkmark \textcircled{1}$$

KCL @ ①

$$4 + I - 12 = 0$$

$$\therefore I = 8A \checkmark \textcircled{1}$$

$$P_2 = +12 \cdot 2I$$

$$= 12 \cdot 2 \cdot 8$$

$$P_2 = +192W \checkmark \textcircled{1}$$

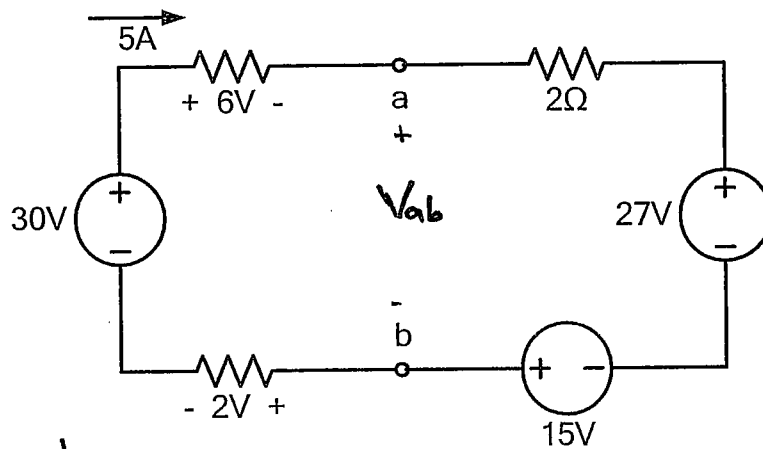
VRAAG/QUESTION 2 [15]

Vraag 2.1 [2]

Bereken V_{ab} en die drywing ge-assosieer met die 15V bron.

Question 2.1 [2]

Calculate V_{ab} and the power associated with the 15V source.



KVL to left of a-b

$$-30 + 6 + V_{ab} + 2 = 0$$

$$-22 + V_{ab} = 0$$

$$\therefore V_{ab} = \underline{22V} \quad \textcircled{1}$$

or

KVL to Right of a-b.

$$(2 \cdot 5) + 27 - 15 - V_{ab} = 0$$

$$10 + 27 - 15 = V_{ab}$$

$$\therefore V_{ab} = \underline{22V}$$

$$P_{15V} = -1 \cdot V$$

$$= -5 \cdot 15$$

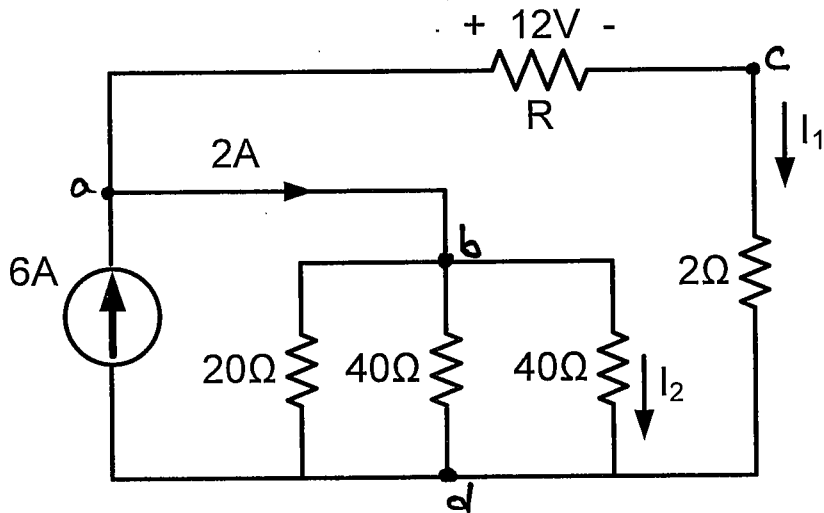
$$P_{15V} = \underline{-75W} \quad \textcircled{1}$$

Vraag 2.2 [3]

Bereken I_1 , I_2 , and R .

Question 2.2 [3]

Determine I_1 , I_2 and R .



KCL @ a : $6 - 2 - I_1 = 0$

$I_1 = 6 - 2 = 4A$ ✓ ①

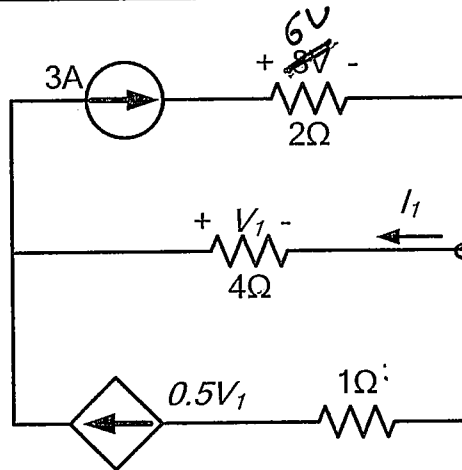
Current divider

$$I_2 = \frac{(20 \parallel 40)}{(20 \parallel 40) + 40} \cdot 2 = \frac{40/3 \cdot 2}{\frac{40}{3} + 40} = \frac{1}{2} = 0.5A$$
 ✓ ①

$V_R = I_1 \cdot R \Rightarrow R = \frac{V_R}{I_1} = \frac{12}{4} = 3\Omega$ ✓ ①

Vraag 2.3 [2]
Bepaal I_1 en V_1 .

Question 2.3 [2]
Determine I_1 and V_1 .



$$3 - I_1 - 0.5V_1 = 0$$

$$3 - I_1 - 0.5(-4I_1) = 0$$

$$3 - I_1 + 2I_1 = 0$$

$$\underline{I_1 = -3A} \quad (1)$$

$$V_1 = -I_1 \cdot 4$$

$$= -(-3) \cdot 4$$

$$\underline{V_1 = 12V} \quad (1)$$

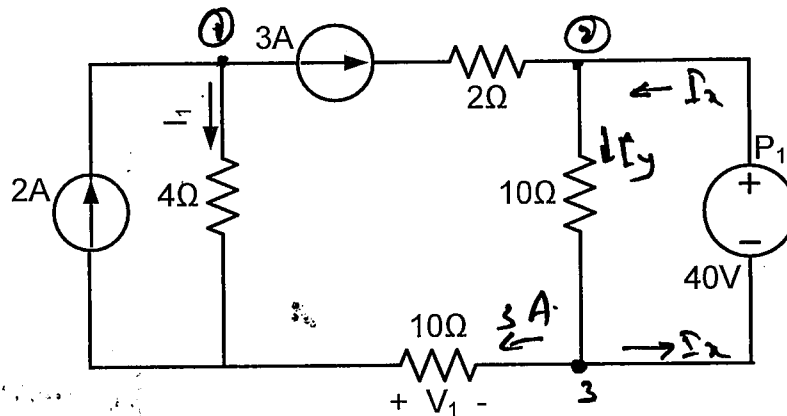
$$V_1$$

Vraag 2.4 [3]

Bereken I_I , V_I en P_I .

Question 2.4 [3]

Calculate I_I , V_I and P_I .



KCL @ ①

$$2 - 3 - I_1 = 0$$

$$\therefore I_1 = -1A \quad \textcircled{1}$$

$$V_I = -3 \cdot 10 = -30V \quad \textcircled{1}$$

up
how

$$\text{KCL @ ②: } 3 + I_x - I_y = 0 \Rightarrow I_y - I_x = 3$$

$$\text{KCL @ ③: } I_y - I_x - I_2 = 0$$

$$\therefore I_2 = I_y - I_x$$

$$\therefore I_2 = 3A$$

$$P_I = -I_x \cdot 40$$

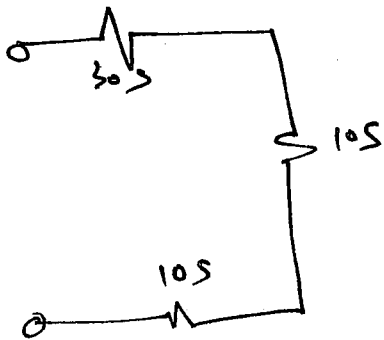
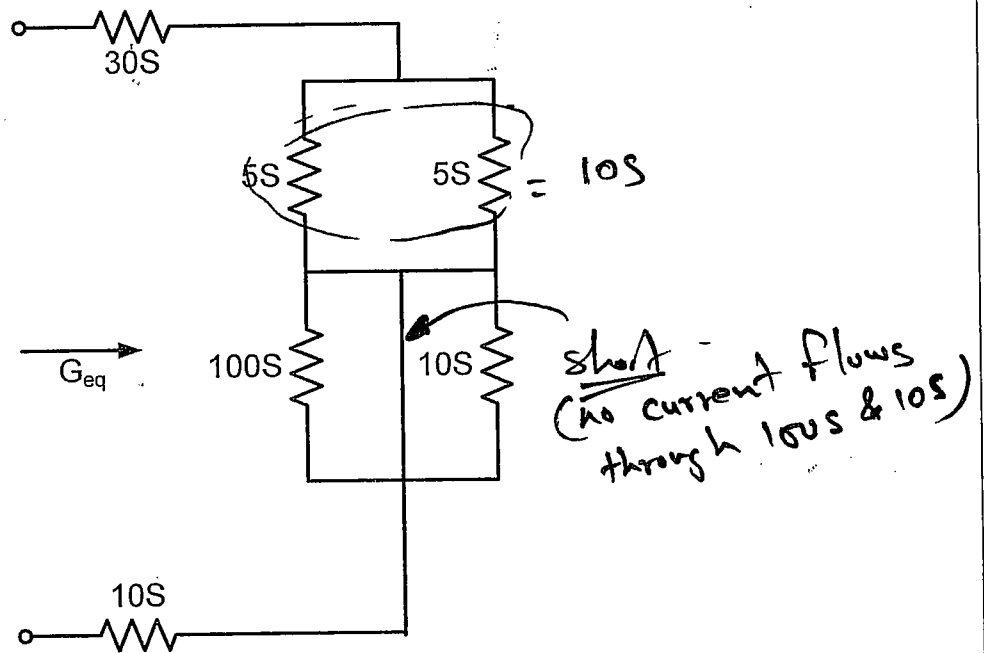
$$I_y = \frac{40}{10} = 4A$$

$$\therefore I_x = 4 - 3 = 1A$$

$$\therefore P_I = -1 \cdot 40 = -40W \quad \textcircled{1}$$

Vraag 2.5 [2]
Bepaal G_{eq} .

Question 2.5 [2]
Find G_{eq} .

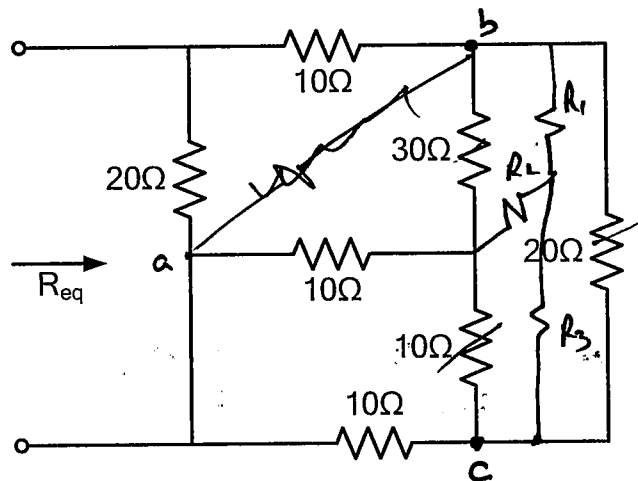


$$G_{eq} = 30 \parallel 10 \parallel 10$$

$$= 30 \parallel 5 = \frac{5 \cdot 30}{5 + 30} = \frac{30}{7}$$

$$G_{eq} = 4.3 \text{ S} \quad \checkmark \quad (2)$$

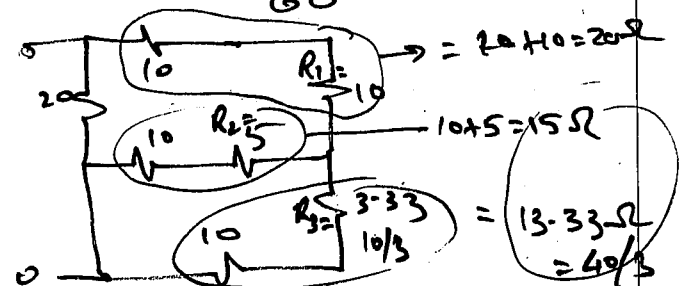
Question 2.6 [3]
Find R_{eq} .

 $\Delta - Y$

$$R_1 = \frac{30 \times 20}{30 + 20 + 10} = \frac{600}{60} = 10 \Omega$$

$$R_2 = \frac{30 \cdot 10}{60} = 5 \Omega$$

$$R_3 = \frac{10 \cdot 20}{60} = 3.33 \Omega = 10/3$$



$$15 \parallel 13.33 = 7.1 \Omega = 120/17$$

$$20 + 7.1 = 27.1 \text{ L}, 20 + \frac{120}{17} = 460/17$$

$$\therefore R_{eq} = 20 \parallel 27.1 = 11.5 \Omega$$

$$\frac{23}{2}$$

③

VRAAG/QUESTION 3 [10]

Vraag 3.1 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$aV_1 + bV_2 = 60$$

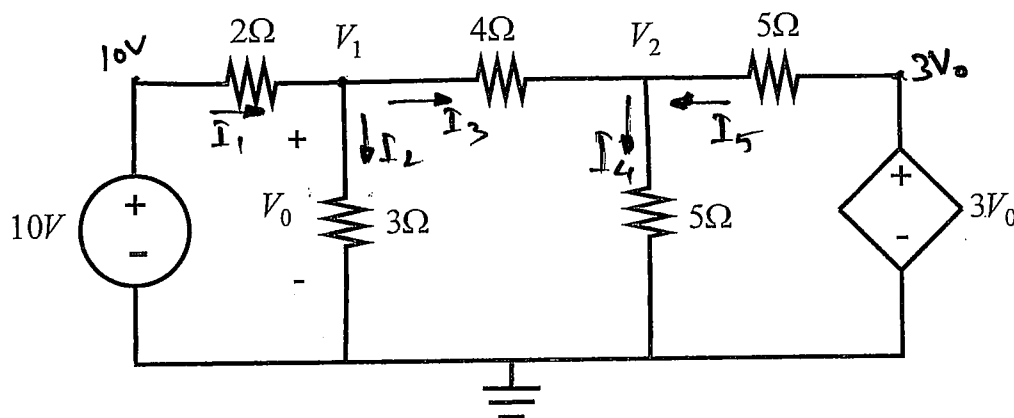
$$cV_1 + dV_2 = 0$$

Question 3.1 [4]

Use nodal analysis to formulate two equations of the following form:

$$aV_1 + bV_2 = 60$$

$$cV_1 + dV_2 = 0$$



KCL @ V_1 :

$$I_1 - I_2 - I_3 = 0$$

$$\frac{10 - V_1}{2} - \frac{V_1}{3} - \frac{V_1 - V_2}{4} = 0$$

$$60 - 6V_1 - 4V_1 - 3V_1 + 3V_2 = 0$$

$$60 - 13V_1 + 3V_2 = 0$$

$$\therefore 13V_1 - 3V_2 = 60 \quad \text{--- (1)}$$

KCL @ V_2 :

$$I_3 - I_4 + I_5 = 0$$

$$\frac{V_1 - V_2}{4} - \frac{V_2}{5} + \frac{3V_0 - V_2}{5} = 0$$

$$5V_1 - 5V_2 - 4V_2 + 12V_0 - 4V_2 = 0$$

$$5V_1 - 13V_2 + 12V_0 = 0$$

$$\text{but } V_0 = V_1$$

$$\therefore 5V_1 - 13V_2 + 12V_0 = 0$$

$$17V_1 - 13V_2 = 0 \quad \text{--- (2)}$$

$$\text{or } -17V_1 + 13V_2 = 0$$

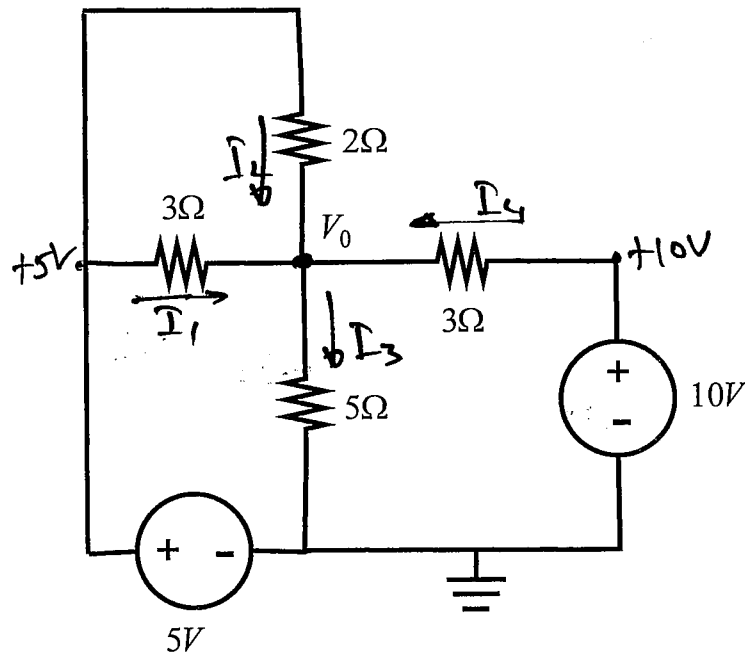
$$\textcircled{4} \begin{bmatrix} 13 & -3 \\ -17 & 13 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \end{bmatrix}$$

Vraag 3.2 [2]

Gebruik node-analise en bereken V_o .

Question 3.2 [2]

Use nodal analysis to calculate V_o .



KCL @ V_o :

$$I_1 + I_2 - I_3 + I_4 = 0$$

$$\frac{5 - V_o}{3} + \frac{5 - V_o}{2} - \frac{V_o}{5} + \frac{10 - V_o}{3} = 0$$

$$50 - 10V_o + 75 - 15V_o - 6V_o + 100 - 10V_o = 0$$

$$225 - 41V_o = 0$$

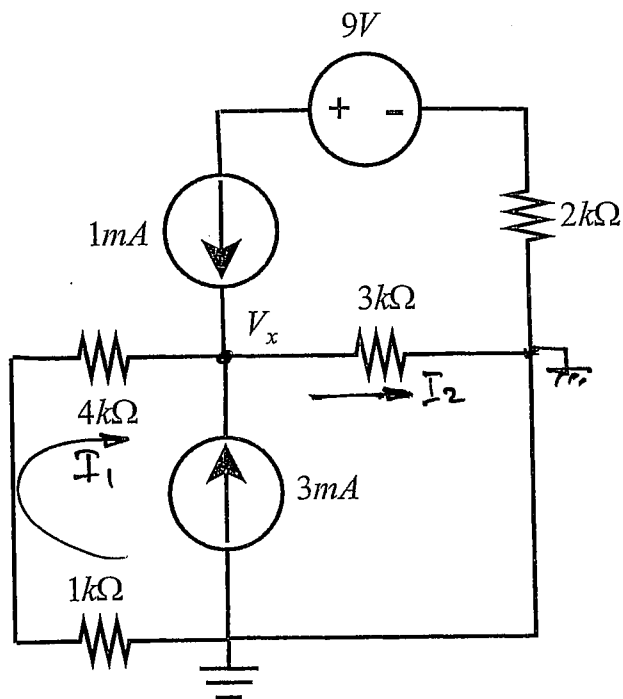
$$\therefore V_o = \frac{225}{41} = \underline{5.5V} \quad \checkmark \quad \textcircled{2}$$

Vraag 3.3 [2]

Gebruik node-analise en bereken V_x .

Question 3.3 [2]

Use nodal analysis to calculate V_x .



$$I_1 + 1m + 3m - I_2 = 0$$

$$I_1 + 1m + 3m - \frac{V_x}{3k} = 0$$

$$\frac{0 - V_x}{4k + 1k} + 4m - \frac{V_x}{3k} = 0$$

$$\frac{-V_x}{5k} - \frac{V_x}{3k} + 4m = 0$$

$$\frac{3V_x + 5V_x}{15k} = 4m$$

$$8V_x = 15k \cdot 4m$$

$$\therefore V_x = \frac{60}{8} = 7.5V$$

Vraag 3.4 [2]Los die matriksvergelyking op om v_2 te bepaal.

$$\begin{bmatrix} 3 & 0 & -1 \\ -1 & 3 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 10 \end{bmatrix}$$

Question 3.4 [2]Solve the matrix equation to find v_2 .

$$\begin{bmatrix} 3 & 0 & -1 \\ -1 & 3 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 10 \end{bmatrix}$$

$$\Delta_2 = \begin{vmatrix} 3 & 10 & -1 \\ -1 & 0 & 0 \\ 0 & 10 & 0 \end{vmatrix} = 10$$

$$\Delta = \begin{vmatrix} 3 & 0 & -1 \\ -1 & 3 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 1$$

$$(0) + (-1 \cdot 1 \cdot -1) + (0) - (0) - (0) - (0) = 1$$

$$\therefore v_2 = \frac{\Delta_2}{\Delta} = \frac{10}{1} = 10 \checkmark \checkmark \textcircled{2}$$

$$v_1 = 30$$

$$v_2 = 10$$

$$v_3 = 80$$

VRAAG/QUESTION 4 [8]

Vraag 4.1 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$7i_1 + ei_2 = f$$

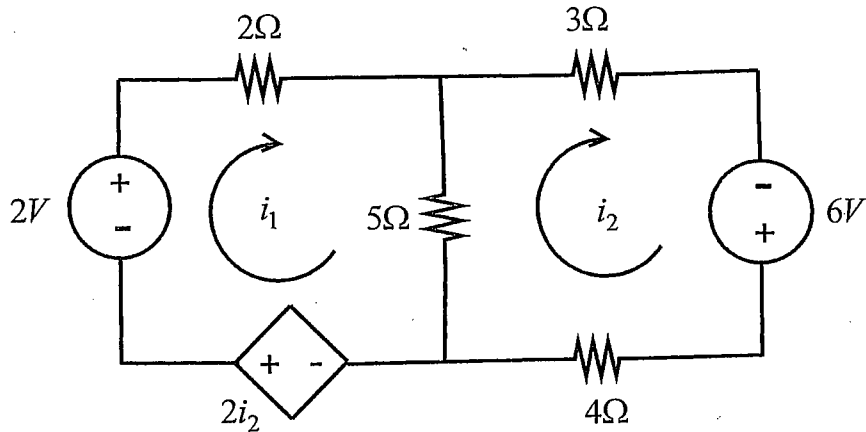
$$gi_1 + hi_2 = 6$$

Question 4.1 [4]

Use mesh analysis to formulate two equations of the following form:

$$7i_1 + ei_2 = f$$

$$gi_1 + hi_2 = 6$$



M1:

$$(2+5)i_1 - 5i_2 = 2 - 2i_2 = 0$$

$$7i_1 - 7i_2 = 2 \quad \text{--- (1)}$$

$$e = -7 \quad f = 2$$

✓

(1)

M2:

$$(3+4+5)i_2 - 5i_1 - 6 = 0$$

$$-5i_1 + 12i_2 = 6 \quad \text{--- (2)}$$

$$g = -5 \quad h = 12$$

✓

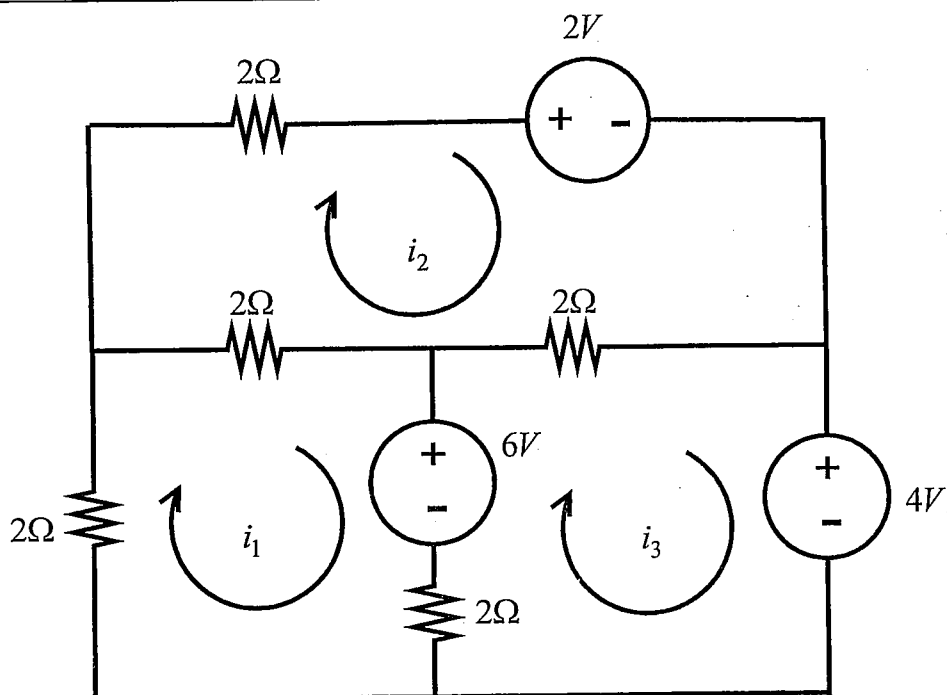
(2)

Vraag 4.2 [3]

Stel 'n matriksvergelyking dmv inspeksie op in terme van die lusstrome.

Question 4.2 [3]

Use inspection to set up a matrix equation in terms of the mesh currents.



$$\begin{bmatrix} (2+2+2) & -2 & -2 \\ -2 & (2+2+2) & -2 \\ -2 & -2 & (2+2+2) \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -6 \\ -2 \\ 2 \end{bmatrix}$$

✓
✓
✓

3

Vraag 4.4 [1]Los die matriksvergelyking op om i_2 te bepaal.

$$\begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Question 4.4 [1]Solve the matrix equation to find i_2 .

$$\begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\Delta = (2 \cdot 4) - (-1 \cdot -1)$$

$$= 8 - 1 = 7.$$

$$\Delta_2 = \begin{bmatrix} 2 & 4 \\ -1 & 5 \end{bmatrix} = (2 \cdot 5) - (4 \cdot -1)$$

$$= 10 - (-4)$$

$$= 14$$

$$\therefore i_2 = \frac{\Delta_2}{\Delta} = \frac{14}{7} = \underline{2 \text{ A}} \quad \text{①}$$